

ROI ET, Thailand

WMO: 484050

Lat: 16.052N

Lon: 103.664E

Elev: 142

StdP: 99.63

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.5	15.9	8.6	7.0	19.0	10.1	7.8	19.5	6.0	24.4	5.4	25.0	0.9	30	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	37.8	25.1	36.7	25.1	35.7	25.2	27.5	32.4	27.1	32.1	26.9	31.7	2.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.3	29.0	26.1	21.9	28.9	25.7	21.4	28.7	88.2	32.4	86.8	32.3	85.5	31.8	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.0	4.3	3.8	DB	12.1	39.3	1.6	1.0	11.0	40.0	10.1	40.6	9.2	41.1	8.1	41.8	(4)
(5)			WB	9.7	28.5	1.7	0.9	8.5	29.2	7.5	29.7	6.5	30.2	5.3	30.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	24.1	26.3	28.8	30.3	29.9	29.5	28.8	28.4	28.1	27.5	26.0	24.0
	DBStd	2.93	2.62	3.08	2.91	2.29	1.71	1.40	1.46	1.25	1.20	1.44	2.14	2.66
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	1	1	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6442	438	456	582	610	617	587	583	572	542	541	480	435
	CDD18.3	3402	180	223	323	360	358	337	325	313	292	283	230	177
	CDH23.3	38405	1644	2396	3976	4718	4483	4164	3764	3532	3136	2903	2148	1540
CDH26.7	17074	639	1136	2107	2603	2227	1924	1547	1337	1092	1073	836	554	
Wind	WSAvg	1.1	1.1	1.0	1.0	1.0	1.0	1.3	1.4	1.3	0.8	0.9	1.2	1.3
Precipitation	PrecAvg	1461	4	19	32	88	209	198	219	272	308	98	14	2
	PrecMax	1866	41	100	134	180	483	296	407	457	581	224	64	25
	PrecMin	1044	0	0	0	10	56	90	96	48	122	19	0	0
	PrecStd	223	10	29	36	51	111	60	83	122	122	58	20	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	36.6	38.7	39.8	38.2	36.3	35.6	34.2	33.7	33.7	34.5	33.7
		MCWB	22.8	23.4	23.9	25.5	26.4	26.3	26.3	26.5	26.6	25.7	25.2	23.7
	2%	DB	32.9	35.4	37.4	38.2	36.7	35.1	34.2	33.2	32.7	32.7	33.2	32.4
		MCWB	22.6	23.6	24.0	25.4	26.3	26.2	26.2	26.3	26.3	25.4	24.8	23.4
	5%	DB	31.7	34.2	36.2	37.0	35.4	34.1	33.1	32.4	32.0	32.0	32.1	31.2
		MCWB	22.4	23.4	24.3	25.5	26.2	26.2	26.0	26.1	26.1	25.1	24.2	22.8
	10%	DB	30.3	32.7	34.9	35.6	34.1	33.2	32.2	31.6	31.2	31.3	31.0	29.9
		MCWB	21.8	23.3	24.3	25.5	26.0	26.1	25.9	26.0	25.8	24.9	23.6	22.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	26.0	26.5	27.4	27.8	27.8	27.5	27.6	27.6	27.0	26.7	25.2
		MCDB	30.9	33.4	33.5	34.8	33.9	32.8	31.9	31.4	31.1	31.0	31.8	31.0
	2%	WB	24.2	25.2	25.9	26.7	27.2	27.2	27.0	27.1	27.1	26.5	25.9	24.3
		MCDB	30.2	32.3	33.5	34.5	33.3	32.3	31.5	31.0	30.7	30.4	31.1	30.3
	5%	WB	23.4	24.6	25.5	26.3	26.9	26.9	26.6	26.7	26.7	26.1	25.3	23.7
		MCDB	29.4	31.4	33.0	33.9	32.9	31.8	31.0	30.6	30.2	29.9	30.0	29.6
	10%	WB	22.5	24.0	25.0	26.0	26.5	26.6	26.3	26.5	26.4	25.7	24.7	22.9
		MCDB	28.8	30.3	32.5	33.4	32.2	31.3	30.6	30.2	29.7	29.4	29.1	28.7
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.0	11.0	9.9	8.2	7.1	6.4	6.1	6.1	7.4	9.4	10.9
		MCDBR	12.7	12.6	12.1	11.2	9.7	8.3	7.7	7.2	7.1	8.2	10.1	11.4
		MCWBR	4.2	3.6	3.1	2.6	2.3	2.1	2.2	2.3	2.3	2.6	3.3	3.8
	5% WB	MCDBR	11.0	10.9	10.6	10.1	8.6	7.4	6.9	6.7	6.6	7.1	8.8	10.5
		MCWBR	4.2	3.5	3.2	2.9	2.6	2.4	2.4	2.5	2.4	2.5	3.2	3.9
Clear-Sky Solar Irradiance	taub	0.477	0.595	0.664	0.620	0.514	0.470	0.460	0.464	0.507	0.525	0.471	0.465	
	taud	2.084	1.788	1.678	1.815	2.137	2.279	2.308	2.287	2.155	2.073	2.197	2.147	
	Ebn at Noon	808	729	695	727	798	824	835	839	806	779	810	807	
	Edh at Noon	156	220	252	220	157	135	131	136	154	163	139	144	
All-Sky Solar Radiation	RadAvg	4.94	5.21	5.29	5.70	5.57	5.21	4.89	4.75	4.65	5.04	4.95	4.77	
	RadStd	0.28	0.26	0.23	0.25	0.30	0.26	0.36	0.31	0.35	0.34	0.18	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.25	N/A	N/A	N/A	+0.43	+0.37	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.27	N/A	N/A	N/A	+0.15	+0.13	N/A	N/A	N/A	N/A

Nomenclature: See separate page